Cloud Based File Security System with Hybrid Encryption

PPD-II Third Review | PID: 23CSEP03

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Outline

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2. Testing
3. Results
4. Analysis
5. Innovation & Technical Contribution
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Implementation Completion

System Overview — CloudVault

- A cloud file storage system with hybrid encryption (AES + RSA)
- Files encrypted on server during upload using AES-256
- Files decrypted **in the browser** during download using Web Crypto API
- User's RSA private key never leaves the browser
- **Zero-effort sharing**: server re-wraps the AES key for the recipient

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Implementation Status — All Features Complete

Completed Modules

Module	Status	Stack
User Authentication	✓	Spring Security + JWT + BCrypt
RSA Key Generation	✓	Web Crypto API + IndexedDB
File Upload & Encryption	✓	AES-256-GCM + RSA key wrapping
File Download & Decryption	✓	Client-side RSA + AES decrypt
File Deletion	✓	Owner-only, disk + DB cleanup
File Sharing	✓	Server-side key re-encryption

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Sharing Module — Newly Completed

Backend (ShareController + ShareService)

- **Share**: Server decrypts AES key using its RSA private key, re-encrypts with recipient's RSA public key, stores SHARED key entry
- **Revoke**: Deletes the recipient's key entry from database
- **List Shared Users**: Returns all users a file is shared with

Frontend (ShareModal + Dashboard)

- Share modal with username input and one-click sharing
- Live list of shared users with revoke buttons
- "Shared With Me" tab showing files from other users
- "Shared by" label on received files

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Testing

Testing Methodology

Manual Functional Testing

- Each feature tested end-to-end through the browser UI
- Backend API verified using browser DevTools network tab
- Cross-user sharing tested with multiple registered accounts

Security Testing

- Verified encrypted files on disk are **completely unreadable** without AES key
- Confirmed RSA private key exists **only in browser IndexedDB**
- Tested unauthorized API access $\rightarrow$ returns 401
- Verified JWT expiry and token validation

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Results

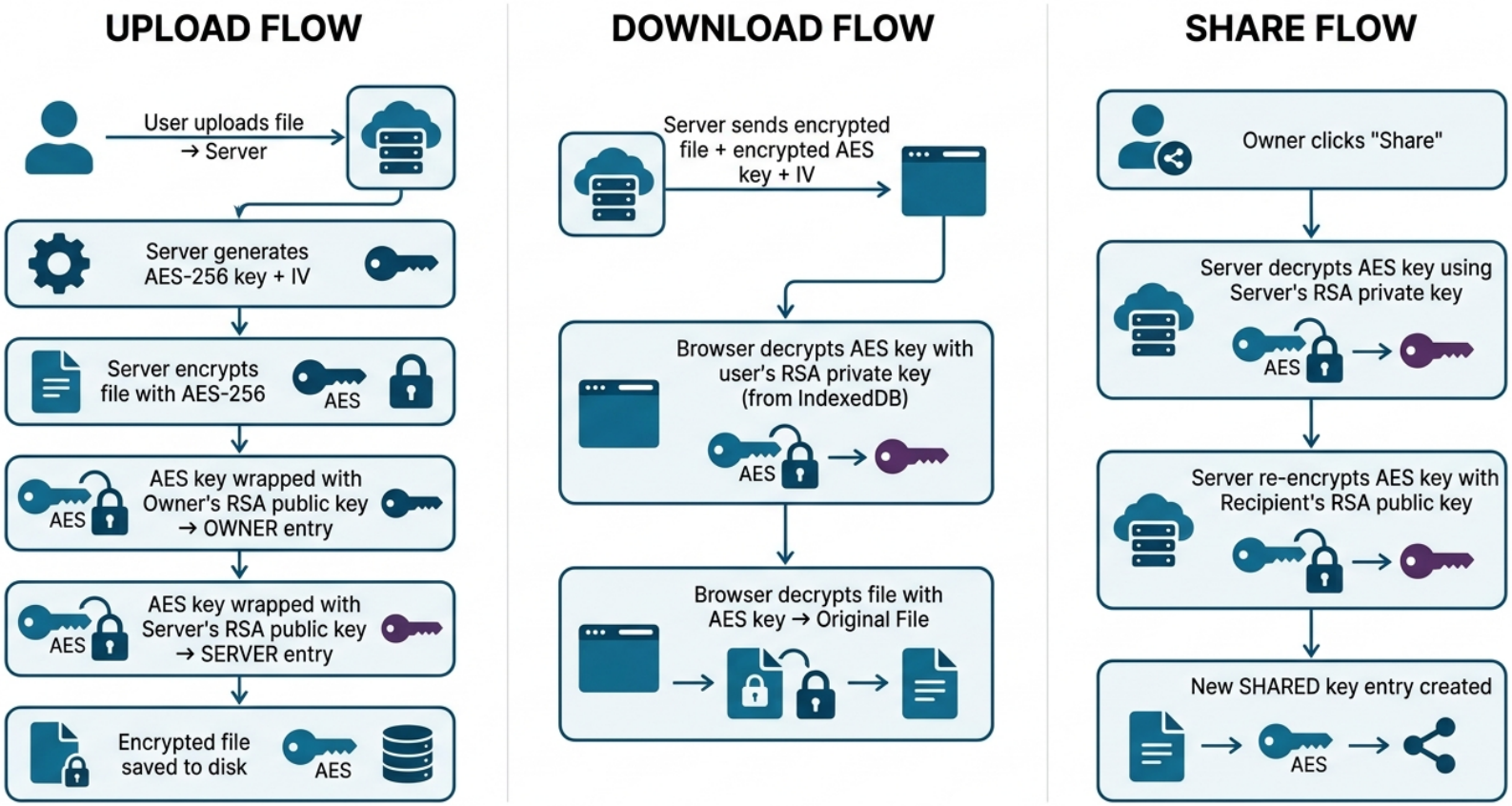
Functional Test Results

All Test Cases Passed

Test Case	Status	Observation
User Registration (RSA keygen)	✓	Key pair generated in browser
User Login (JWT flow)	✓	Token stored, API access works
File Upload (AES encryption)	✓	Encrypted file stored on disk
File Download (client decrypt)	✓	Original file recovered
File Deletion (owner only)	✓	Disk + DB cleaned up
Share file with user	✓	Recipient sees file in shared tab
Download shared file	✓	Decrypts with recipient's key
Revoke shared access	✓	File removed from shared tab
Unauthorized access	✓	Returns 401, blocks access

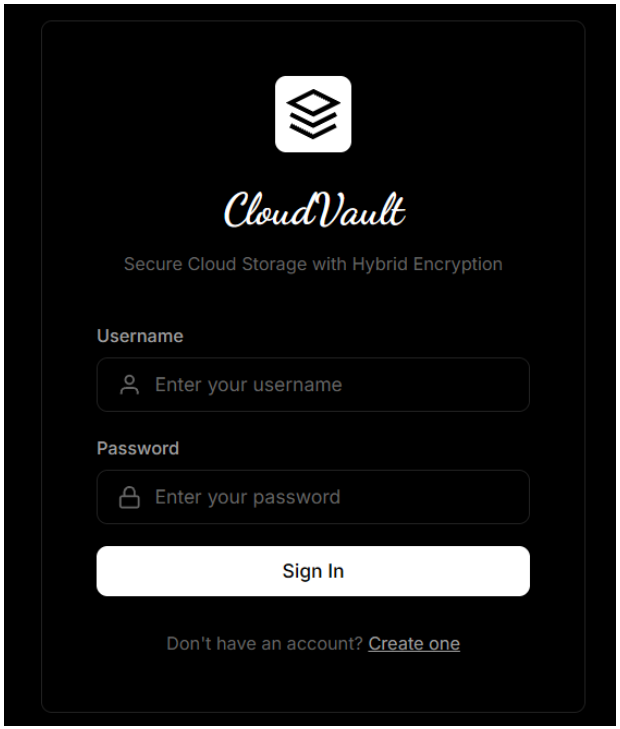
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Encryption Workflow

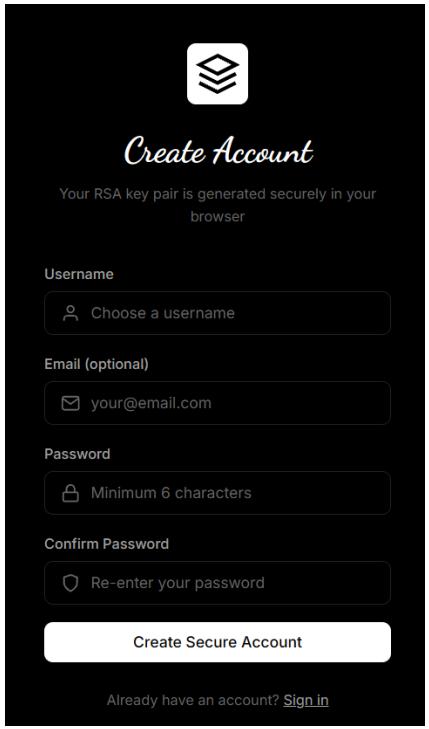


Demo — Login & Registration

Login Page

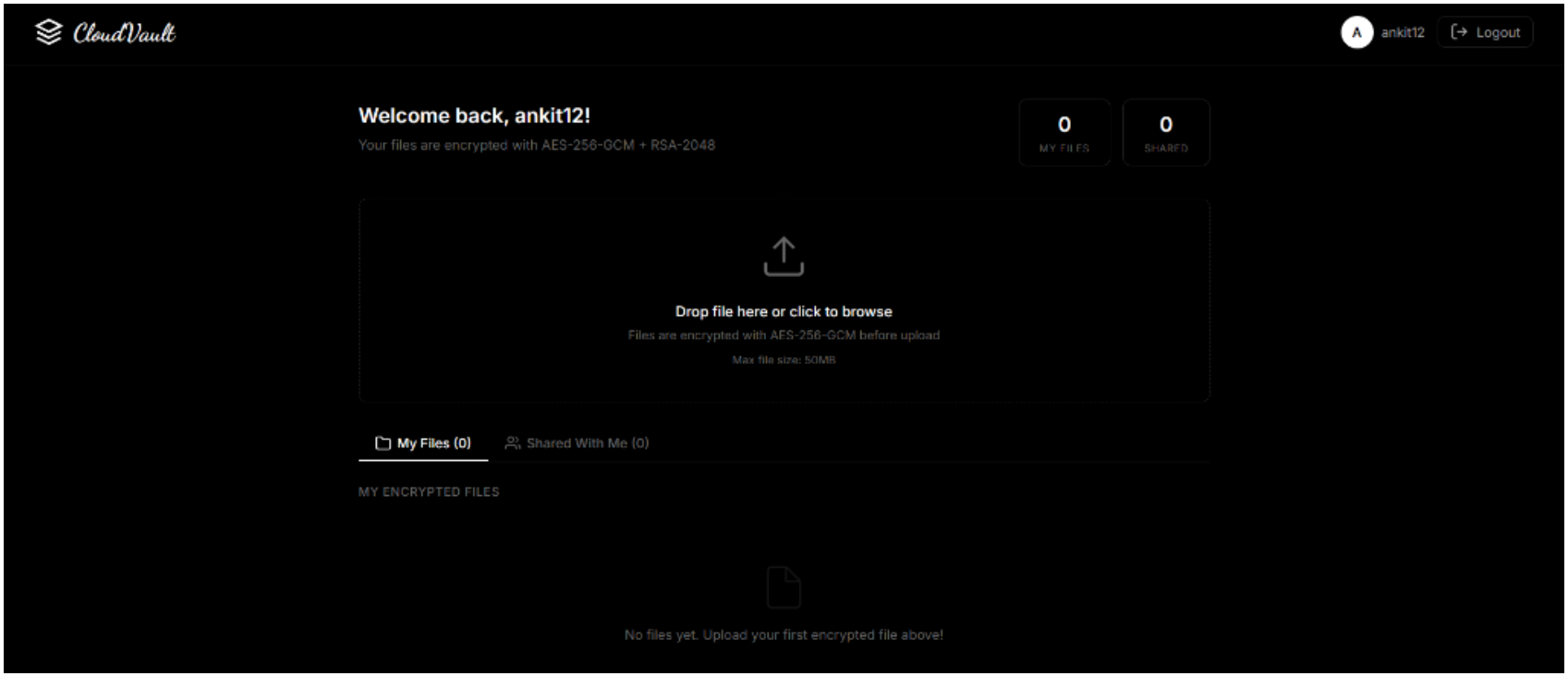


Register Page



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Demo — Dashboard



Analysis

Performance Analysis

Measured Results

Operation	Performance
RSA-2048 key generation (browser)	Under 1 second
File encryption (under 50 MB)	Near-instantaneous
Client-side AES decryption	Near-instantaneous
RSA key unwrapping	Under 100 ms
File sharing (key re-encryption)	Under 200 ms
JWT authentication overhead	Negligible

Observation

Hybrid encryption adds minimal overhead — the AES key is only 32 bytes, so RSA operations on it are extremely fast.

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Security Analysis

Multi-Layer Security

Layer	**Mechanism**
Passwords	BCrypt hashing with random salt
Sessions	Stateless JWT tokens (24h expiry)
API Protection	Spring Security filter + JWT validation
Files at Rest	AES-256-GCM encrypted on disk
Key Protection	RSA-wrapped AES keys per user
Client Keys	RSA private key in browser IndexedDB
Server Keys	RSA key pair auto-generated at startup

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Threat Analysis

Threats Mitigated

- **Server breach**: Attacker gets encrypted files + wrapped keys → cannot decrypt without user's RSA private key (stored only in browser)
- **Network interception**: JWT auth + encrypted file transfer → raw file never travels over the network
- **Unauthorized access**: Spring Security blocks all unauthenticated API calls

Known Limitation

Clearing browser data deletes the RSA private key. A key export/backup feature is recommended for production.

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Innovation & Technical Contribution

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Innovation — What Makes CloudVault Unique

Key Innovations

1. **Browser-Only Private Keys**: RSA private key is generated and stored exclusively in the browser (Web Crypto API + IndexedDB). The server never sees it.
2. **Zero-Effort Sharing**: Files are shared without re-encrypting the entire file — only the AES key is re-wrapped using the recipient's RSA public key.
3. **Hybrid Encryption with Key Envelope**: Each file has per-user encrypted key entries (OWNER / SERVER / SHARED), enabling granular access control.
4. **Client-Side Decryption**: The original file is reconstructed entirely in the browser — the server only delivers ciphertext.

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Technical Contribution

Contributions to Cloud Security

- Demonstrated a **practical hybrid encryption model** for web-based file storage using standard browser APIs
- Implemented a **Key Envelope Table** design pattern for multi-user encrypted file access
- Achieved **zero-knowledge file sharing** where the server facilitates sharing
- Built a **full-stack reference implementation** using modern frameworks (Spring Boot 3 + React 19 + Web Crypto API)

Tech Stack

Spring Boot 3.2.5 (Java 17) | React 19 + Vite 8 | Web Crypto API | JWT | AES-256-GCM | RSA

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Thank You

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